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# CYCLE FREQUENCY-HARMONIC-TIME TRANSFORMER FOR NOTE-LEVEL SINGING VOICE TRANSCRIPTION

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## ABSTRACT

Singing voice transcription (SVT) is the task of converting singing voice music into symbolic note series. Although most SVT models utilized the time-frequency information from the input spectrogram, the useful harmonic information in singing voices has not been utilized enough. In this paper, we propose a novel 3D Cycle Frequency-Harmonic-Time Transformer (**CFT**) to explicitly capture the harmonic series of singing voices, where we first define a tokenization scheme that captures harmonics across multiple octaves, then the harmonic features are aggregated into the frequency-harmonic-time representations via a cyclic architecture. Results show that our method achieves state-of-the-art performances on several public datasets, including note-wise accuracy increases of 5.76% for MIR-ST500 and 13.56% for Cmedia.

***Index Terms*—** music information retrieval, automatic music transcription, note-level singing voice transcription

## 1. INTRODUCTION

Singing voice transcription (SVT) is a process that requires the precise modelling of the singing voice, including the quantification of the note's onset, offset, pitch, and generates a series of note sequences as output [1, 2]. It is useful for generative music AI research and has been widely applied in music education [3], singing voice synthesis [4], automatic singing skill evaluation [5], query-by-humming [6], and other relevant music information retrieval (MIR) tasks.

One of the core tasks in SVT is to detect the fundamental frequency ($f_0$) from the input singing voice. From the perspective of music, $f_0$ is the perceived pitch of a musical note that is the first member of the harmonic series. An ideal set of harmonic series ($2*f_0, 3*f_0, 4*f_0, \cdots$) contains positive integer multiples of $f_0$, as shown in Fig. 1. The octave series ($2*f_0, 4*f_0, 8*f_0, \cdots$) is a special subset of the harmonic series where each shares the same pitch class as the $f_0$ and is one or multiple octaves higher than $f_0$. Within each octave (e.g., $[4*f_0, 8*f_0]$), more harmonics (e.g., $5*f_0, 6*f_0$) can be found that span a different musical interval from $f_0$ than an octave, and the whole harmonic series contributes to the pitch, timbre, and loudness that we perceive in singing voices. The harmonic series in different singing voices usually vary, causing the timbre of one voice different from another.

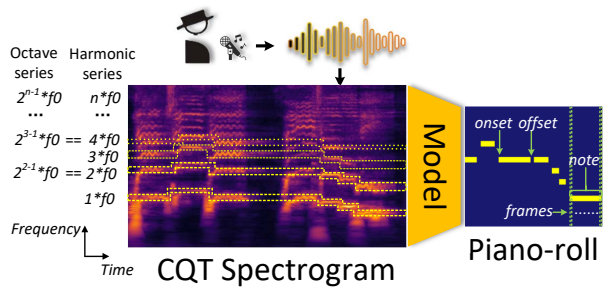


**Fig. 1**. Illustration of the singing voice transcription (SVT) process. The left one represents the constant-Q-transform (CQT) spectrogram extracted from the given singing clip, with the fundamental frequencies $f_0$ and upper harmonics (e.g., $2*f_0, 3*f_0$, and $4*f_0$) highlighted in dot lines. As discussed in Section 1, they form the harmonic series $n*f_0$, and its subset octave series $2^{n-1}*f_0$ that spans different octaves from $f_0$. The right one is the ideal output of SVT containing transcribed notes of the symbolic representation (piano-roll).

Recently, there has been MIR research that utilized the harmonic series of music signals: the Harmonic Constant-Q Transform (HCQT) [7, 8] captures harmonic relationships through a three-dimensional CQT array and applies to pitch estimation in polyphonic music. In piano transcription, Wei et al. [9] designed harmonic dilated convolutions, where the dilation is set manually to obtain specific harmonics. Miguel et al. [10] further proposed the 3D harmonic convolutions to capture multiple harmonics automatically for piano music. Despite recent research that either relied on domain knowledge [11, 12, 13] or used semi-supervised/self-supervised learning to enhance data representation [14, 15] in SVT, the harmonic series in singing voices has not been utilized. Since the core goal of SVT is to identify the correct pitches for singing, and the timbre in singing voices is usually different, it can be beneficial to disentangle pitch from timbre by explicit modelling of harmonic series in singing voices.

In this paper, we propose a transformer-based SVT model called Cycle frequency-Harmonic-Time Transformer (**CFT**)

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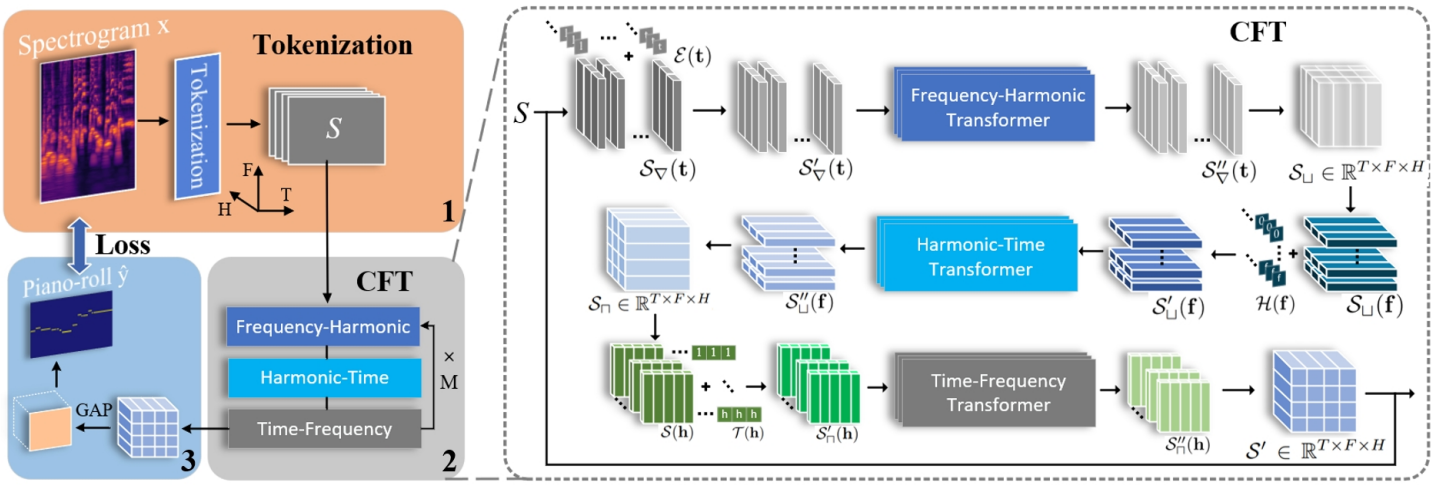


**Fig. 2**. Overview of **CFT**, where the left part is the pipeline and the right part is the cycle structure. The pipeline on the left follows the three steps below: **(1)** The input CQT spectrogram $\mathbf{x} \in \mathbb{R}^{T \times F}$ is transformed as the 3D tokens, represented as $\mathcal{S} \in \mathbb{R}^{T \times F \times H}$ through a tokenization scheme (see Section 2.2), where $T$ is the number of frames, $F$ is the dimension of frequency bins, $H$ is the dimension of harmonic features. **(2)** The **CFT** aggregates sequence-level frequency-harmonic, harmonic-time, and time-frequency features following the cycle structure to construct the 3D frequency-harmonic-time representations $\mathcal{S}' \in \mathbb{R}^{T \times F \times H}$. **M** refers to the number of times the cycle structure is performed. The details of **CFT** shown on the right are introduced in Section 2.3. **(3)** Global average pooling (GAP) along the harmonic axis is used to sum up the harmonic features and generate the predicted piano-roll $\hat{\mathbf{y}} \in \mathbb{R}^{T \times N}$ as the transcription result, where $N$ is the singing voice pitch range represented as MIDI note indexes.

that utilizes harmonic series in singing voices. The proposed model has two components. The first is a tokenization scheme that extracts 3D tokens with time, frequency, and the additional harmonic features from singing voices using the prior information of octave and harmonic series (see Section 2.2). The second is a 3D transformer-based architecture, which constructs a cycle structure that models sequence-level frequency-harmonic, harmonic-time, and time-frequency features to generate the 3D frequency-harmonic-time representations for the comprehensive modelling of singing voices (see Section 2.3). Specially, the harmonic features are aggregated into a cyclic architecture that contains three transformer encoders, where each encoder extracts frequency-harmonic, harmonic-time, and time-frequency features in order, resulting in the 3D frequency-harmonic-time representation to predict the final transcription for singing voices. Results show that **CFT** achieves state-of-the-art (SOTA) performances on several public datasets, including MIR-ST500, Cmedia (both Chinese pop), ISMIR2014 (English pop), and TONAS (Spanish flamenco).

## 2. METHODOLOGY

### 2.1. Overview

The pipeline of **CFT** is shown in Fig. 2. Given an input Constant-Q Transform (CQT) spectrogram of input singing voice music, our model outputs a posteriorgram (piano-roll) as the transcription result. The three steps of **CFT** are highlighted on the left of Fig. 2 in different colours: tokenization (orange), cycle frequency-harmonic-time transformer (grey), and the output piano-roll prediction (blue). The details of tokenization and **CFT** will be introduced in Section 2.2 and Section 2.3, respectively. The goal of **CFT** is to detect the onset/offset and pitch of note series, we define the loss function $\mathcal{L}$ as follows:

$$\mathcal{L} = \sum_{t=1}^{T} \sum_{n=1}^{N} (l_{\mathbf{onset}}^{\hat{\mathbf{y}}(t,n)} + l_{\mathbf{frame}}^{\hat{\mathbf{y}}(t,n)} + l_{\mathbf{offset}}^{\hat{\mathbf{y}}(t,n)}), \tag{1}$$

where $l_{\mathbf{onset}}^{\hat{\mathbf{y}}(t,n)}$, $l_{\mathbf{frame}}^{\hat{\mathbf{y}}(t,n)}$, $l_{\mathbf{offset}}^{\hat{\mathbf{y}}(t,n)}$ are the binary cross-entropy loss between the ground-truth and predictions $\hat{\mathbf{y}}(t,n)$. $t$ denotes the frame-level time stamp and $n$ is the pitch range of the corresponding MIDI index.

### 2.2. Tokenization scheme

Tokenization is the step of converting the raw input into a transformer-readable format. Many existing MIR methods utilize convolutional neural networks (CNNs) as the tokenization scheme for transformer-based models [13, 16, 17, 18]. However, traditional CNNs can only extract local features of spectrograms, making it difficult to capture harmonic series that are often spread out in the spectrogram (see Fig. 1) and leading to compromised SVT performances. To incorporate the information of harmonic series, we adopt the existing 3D

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harmonic convolutions [10] from the perspectives of octaves and pitch class to model the 3D frequency-harmonic-time representations. As discussed in Section 1, the octave series is a special subset of the harmonic series where each shares the same pitch class as the $f_0$ and is one octave higher or lower than its neighbour. Therefore, we first use each member of the octave series to divide the input CQT spectrogram along the frequency axis by octaves, where each frequency bin corresponds to a pitch class, as shown in Fig. 3. Then, the 3D harmonic convolution automatically captures the features of harmonic series features from each octave. Each kernel of 3D harmonic convolution extracts the harmonic features of a channel and sums them up in the output feature maps to obtain the 3D tokens $\mathcal{S} \in \mathbb{R}^{T\times F\times H}$.

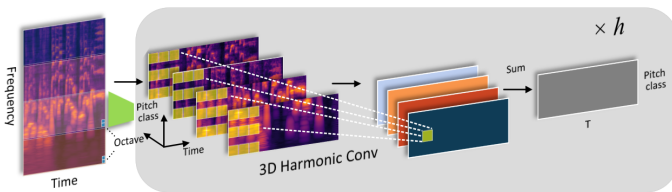


**Fig. 3**. Tokenization scheme. The CQT spectrogram is divided into octave spectrograms along the frequency axis. We use $h$ 3D harmonic convolution kernels to extract $h$ channels of harmonic features for each octave.

### 2.3. Cycle frequency-Harmonic-Time Transformer

Since our tokenization scheme captures additional harmonic features as discussed above, we require a new architecture different from previous transformer-based ones [16, 17] to provide a space to aggregate useful features of time, frequency, and harmonics. Therefore, we build a cycle structure of $frequency \rightarrow harmonic \rightarrow time \rightarrow frequency$ to construct a 3D space for modelling sequence-level frequency-harmonic, harmonic-time, and time-frequency features to generate 3D frequency-harmonic-time representations.

The proposed **CFT** architecture is shown in Fig. 2, which contains three transformer-based models: frequency-harmonic transformer, harmonic-time transformer, and time-frequency transformer. Each of them includes an attention mechanism and a feed-forward network to capture the sequence dependency. In the context of SVT, this process means the interactions among the frequency, harmonic, and time and are explicitly modeled to build the 3D representations of onset/offset and pitch.

**Frequency-Harmonic Transformer** To capture the dependency between frequencies and harmonics, we first partition the input tokens $\mathcal{S} \in \mathbb{R}^{T\times F\times H}$ along the time axis to the 2D frequency-harmonic sequence embeddings $[\mathcal{S}_\triangledown(\mathbf{1}), \mathcal{S}_\triangledown(\mathbf{2}), \ldots, \mathcal{S}_\triangledown(\mathbf{t})] \in \mathbb{R}^{F\times H}$ shown in Fig. 2. Since singing is represented as signals in the time series, each sequence embedding is added with a learnable temporal embedding $\mathcal{E}(\mathbf{t}) \in \mathbb{R}^{F\times H}$ to incorporate the frame-level timestamp into the frequency-harmonic representations:

$$\mathcal{S}'_\triangledown(\mathbf{t}) = \mathcal{S}_\triangledown(\mathbf{t}) \oplus \mathcal{E}(\mathbf{t}), \tag{2}$$

Where $\oplus$ denotes the element-wise addition. The result $\mathcal{S}'_\triangledown(\mathbf{t})$ carries information between the frequency and harmonic to the following transformer encoders that perform $t$ times parallel calculations to obtain the latent sequence embeddings $\mathcal{S}''_\triangledown(\mathbf{t})$. The attention layer in transformer encoders attempts to find the frequency positions corresponding to harmonic information to establish their sequence dependency. Finally, to aggregate the frequency-harmonic features, we compile the $[\mathcal{S}''_\triangledown(\mathbf{1}), \mathcal{S}''_\triangledown(\mathbf{2}), \ldots, \mathcal{S}''_\triangledown(\mathbf{t})] \in \mathbb{R}^{F\times H}$ to frequency-harmonic representations $\mathcal{S}_\sqcup \in \mathbb{R}^{T\times F\times H}$.

**Harmonic-Time Transformer** To capture harmonic and time dependency, we partition the output representations $\mathcal{S}_\sqcup$ to the 2D harmonic-time sequence embeddings $[\mathcal{S}_\sqcup(\mathbf{1}), \mathcal{S}_\sqcup(\mathbf{2}), \ldots, \mathcal{S}_\sqcup(\mathbf{f})] \in \mathbb{R}^{H\times T}$ along the frequency axis. As shown in the original transformer paper [19], a sinusoidal positional encoding was added to the input sequence, so the subsequent layers can recognize the order of input elements. Similarly, we incorporate a learnable frequency-wise positional encoding $\mathcal{H}(\mathbf{f}) \in \mathbb{R}^{H\times T}$ into each sequence embedding to encode the sequence order of the harmonic-time representations:

$$\mathcal{S}'_\sqcup(\mathbf{f}) = \mathcal{S}_\sqcup(\mathbf{f}) \oplus \mathcal{H}(\mathbf{f}), \tag{3}$$

The $f$ stacks of transformer encoders can capture the correlation of time and harmonics within their respective frequency bins to obtain the latent sequence embeddings $\mathcal{S}''_\sqcup(\mathbf{f})$. In the context of SVT, harmonic features and time are closely related to the activated pitch contours, which is crucial for note onset/offset detection. Similarly, we also compile the $[\mathcal{S}''_\sqcup(\mathbf{1}), \mathcal{S}''_\sqcup(\mathbf{2}), \ldots, \mathcal{S}''_\sqcup(\mathbf{f})] \in \mathbb{R}^{H\times T}$ to harmonic-time representations $\mathcal{S}_\sqcap \in \mathbb{R}^{T\times F\times H}$.

**Time-Frequency Transformer** Similar to the two transformers introduced above, we partition the output representations $\mathcal{S}_\sqcap$ along the harmonic axis to the 2D time-frequency sequence embeddings $[\mathcal{S}_\sqcap(\mathbf{1}), \mathcal{S}_\sqcap(\mathbf{2}), \ldots, \mathcal{S}_\sqcap(\mathbf{h})] \in \mathbb{R}^{T\times F}$. We also introduce learnable harmonic-wise positional encoding $\mathcal{T}(\mathbf{h}) \in \mathbb{R}^{T\times F}$ to focus on the sequence order of the time-frequency representations:

$$\mathcal{S}'_\sqcap(\mathbf{h}) = \mathcal{S}(\mathbf{h}) \oplus \mathcal{T}(\mathbf{h}), \tag{4}$$

Then, $\mathcal{S}'_\sqcap(\mathbf{h})$ is fed into transformer encoders to model the time-frequency representations of each harmonic feature and aggregate time-frequency features. It is noteworthy that the time-frequency transformer serves as the final stop in the cyclic structure. By this stage, the **CFT** has accomplished a cycle of modelling for frequency, harmonic, and time features to generate the representations $\mathcal{S}' \in \mathbb{R}^{T\times F\times H}$ as shown in Fig. 2. If there is a desire to increase the depth of the network, one simply needs to repeat the cycle structure.

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| | DATASET | Metrics | Tony [20] | VOCANO [14] | EFN [21] | JDC$_{note}$ [15] | CECTC[22] | MusicYOLO [12] | AV-SVT [13] | **CFT** | $\Delta$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **ID evaluation** | MIR-ST500 | COnPOff | | 21.15 | 45.78 | 42.23 | 57.19 | 57.21 | 52.39 | **57.77** | +0.56 |
| | | COnP | \ | 38.31 | 66.63 | 69.74 | 74.37 | 71.42 | 70.73 | **80.13** | +5.76 |
| | | Con | | 50.47 | 75.44 | 76.18 | 79.62 | 78.17 | 78.32 | **82.51** | +2.89 |
| | Cmedia | COnPOff | | 28.28 | 35.13 | 40.20 | | | | **55.80** | +15.60 |
| | | COnP | \ | 48.33 | 60.77 | 66.11 | \ | \ | \ | **79.67** | +13.56 |
| | | Con | | 64.56 | 76.40 | 75.97 | | | | **82.43** | +6.03 |
| **OD evaluation** | ISMIR2014 | COnPOff | 50.11 | 50.18 | 49.55 | | 63.60 | **69.37** | 52.36 | 63.56 | -5.81 |
| | | COnP | 68.17 | 63.30 | 63.63 | \ | 76.73 | 80.68 | 70.38 | **80.89** | +0.21 |
| | | Con | 73.04 | 77.55 | 79.16 | | **93.01** | 85.18 | 92.7 | 90.15 | -2.86 |
| | TONAS | COnPOff | | | 9.57 | | | | 12.71 | **23.87** | +11.16 |
| | | COnP | \ | \ | 19.65 | \ | \ | \ | 25.24 | **30.90** | +5.66 |
| | | Con | | | 42.41 | | | | 52.77 | **57.81** | +5.04 |

**Table 1**. The transcription results of MIR-ST500, Cmedia, ISMIR2014, and TONAS datasets. The details of datasets and evaluation metrics are introduced in Section 3.1 and Section 3.2, respectively. The horizontal line in between indicates the first two datasets are in-domain (ID) validations, which contain Chinese Pop for both training and testing. the last two datasets are out-of-domain (OD) validations, which are non-Chinese style and used to demonstrate the generalization ability of our proposed **CFT**. The best and the second-best performances are highlighted in bold and underlines, respectively. $\Delta$ represents the performance differences between the proposed **CFT** and the previous SOTAs.

## 3. EXPERIMENTS

### 3.1. Dataset

Our experiments involve the four following datasets, which all contain solo singing and the corresponding vocal note annotations.

- **MIR-ST500** [21] is the largest public dataset used in SVT. It consists of 500 Chinese pop songs.
- **Cmedia** contains 100 Chinese pop songs and has been used in the SVT task from the Music Information Retrieval Evaluation eXchange (MIREX) [1].
- **ISMIR2014** [23] contains 38 English pop vocal clips sung by both untrained adults and children.
- **TONAS** [24] contains 72 flamenco singing excerpts (Deblas, two variants of Martinete).

Following the baseline models [12, 13, 15, 22, 21], we use the same 400 songs of MIR-ST500 for training and the remaining 100 songs for testing. Cmedia, ISMIR2014, and TONAS datasets are used for testing only.

### 3.2. Evaluation Metrics

Following previous work [12, 13, 15, 22, 21], we used the following evaluation metrics: `COn` (Correct onset F1 score) evaluates whether the onset of notes is consistent with the ground truth. `COnP` (Correct onset and pitch F1 score) considers a prediction correct if it has the same pitch as a reference onset. `COnPOff` (Correct onset pitch and offset F1 score) extends the previous metrics by additionally considering note offset. All metrics are calculated using $mir_eval$ [25] to obtain the average F-score. We set the onset threshold at 50 ms and the offset threshold at 0.2$\times$ note duration.

[1] https://www.music-ir.org/mirex/wiki/2020:Singing_Transcription_from_Polyphonic_Music

### 3.3. Implementation Details

We use a batch size of 16 and adopt the Adam [26] optimizer with a learning rate of 3e-4 for training. The input audio is resampled to 16 kHz, then a 288-bin CQT spectrogram is computed, which represents 72 distinct semitones from G1 to F7. Regarding tokenization, it used kernels of size (4, 3/5/7, 3)$\in$(octave, pitch class, time) to span the input spectrogram across six octaves, extracting harmonic features in each octave. For the **CFT**, in each transformer, we use only one transformer encoder to capture the sequence dependency in harmonics, time, and frequency.

### 3.4. Baselines

We use multiple existing SVT models as baselines to compare with our proposed model, and we briefly introduce each one as follows: Tony [20] is a free, open-source note-level transcription software that uses both the pYIN algorithm [27] and a Hidden Markov Model to quantify note-level transcription results. VOCANO [14] is a semi-supervised framework with virtual adversarial training. JDC$_{note}$[15] uses extracted pseudo-labels to train a noisy teacher-student SVT framework. EFN [21] is a model based on the image classification model Efficient-Net and trained using the MIR-ST500 dataset. CECTC [22] uses CTC loss with cross-entropy loss to optimize the transcription model. AV-SVT [13] introduced

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multimodal learning for singing voice transcription. MusicYOLO [12] is based on the object detection model YOLOX, and note-level transcription is completed by a spectrogram peak search.

### 3.5. Results

We use the largest dataset MIR-ST500 for training and testing our model. The results are shown in Table 1, where we consider `COnP` as the most critical metrics, representing the correctness of the note's onset and pitch [17, 13]. Additionally, we use MIR-ST500 and Cmedia for in-domain (ID) evaluations, because MIR-ST500 is used for training and they both focus on the Chinese Pop style. ISMIR2014 are TONAS are used for out-of-domain (OD) evaluation since the former is English Pop and the latter is Spanish Flamenco, which differs drastically from the Chinese Pop style. Among all the evaluation metrics shown in Section 3.2.

For the ID evaluation, **CFT** outperforms the previous SOTA metrics by 5.76% (MIR-ST500) and 13.56% (Cmedia) respectively in the core metric `COnP`. In addition, CFT even outperforms the Time-Frequency Perceiver[2] [17] on MIR-ST500 by 1.6% (80.13% VS. 78.5% COnP), where the latter is trained with a significantly larger collection of data involving 4044 songs of both instrument and singing annotations. The superior results confirm that our method, which utilizes the essential characteristics of harmonics for singing notes, is effective and can achieve a higher transcription performance than other baselines.

For the OD evaluation, **CFT** achieves 80.89% `COnP` on the ISMIR2014 dataset, slightly higher than the previous SOTA MusicYOLO. **CFT** has surpassed all baselines on the metrics `COn`, `COnP`, `COnPOff` for the TONAS dataset. The superior performances on out-of-domain datasets indicate the strong robustness of the **CFT**, proving that the designed 3D frequency-harmonic-time representation effectively utilizes the sequence-level modelling and generates the more generic representation of singing voices.

### 3.6. Ablation Study

To understand the significance of each modification in our model, we conduct an ablation study shown in Fig. 4. The curve of the "proposed model" demonstrates that **CFT** effectively utilizes the musical harmonic series extracted by tokenization through consistent modelling of sequence-level frequency-harmonic-time representations. The curves of the "wo time/frequency/harmonic" show that the lack of harmonic modelling has the biggest impact among them. This proves the necessity of harmonic series for SVT tasks and the effective modelling of harmonic features by the proposed **CFT**. The performance of the "wo tokenization" is better than the proposed **CFT** at iterations < 20000, as pre-defined harmonic distances allow the model to capture harmonic features with less training. However, as the training process continues, the proposed tokenization can automatically capture the harmonics of different octaves, leading to better results.

[2]We did not include this work as our baseline since it focused on multi-track music transcription, which is essentially a different task than SVT.

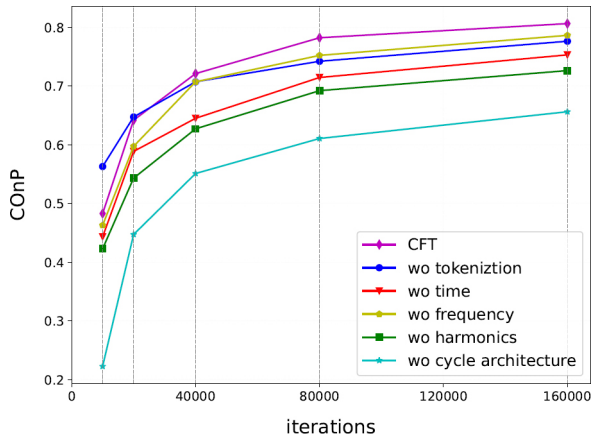


**Fig. 4**. Ablation study on the MIR-ST500 test set with the metric `COnP`. "**CFT**" represents our proposed method. "wo time/frequency/harmonics" means the proposed **CFT** omits any one of the dimensions among time, frequency, and harmonics. "wo tokenization" refers to the use of 2D harmonic convolution [9] that contains pre-defined harmonic distances to extract harmonic features instead of our tokenization scheme. "wo cycle architecture" refers to using the three original transformers [19] instead of **CFT**.

## 4. CONCLUSION

In this paper, we propose a novel note-level singing voice transcription method, namely the Cycle Frequency-Harmonic-Time Transformer (**CFT**). **CFT** that establishes a 3D tokenization scheme to extract tokens with harmonic series and builds a 3D singing voice representation space through cycle sequence-level modelling of the frequency-harmonic-time representations. Experimental results show that our proposed method achieves state-of-the-art performances on both in-domain and out-of-domain evaluations, demonstrating the effectiveness and robustness of **CFT**.

Since the training dataset in this paper only contains Chinese pop music, we will extend our model by training with singing annotations from various genres and languages. In the future, we will also apply this method to multi-track music transcription, enabling our model to transcribe not only singing but also other instrument sources from real-world recordings in an end-to-end manner.

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